1. Let $f(n) = \sum_{y=1}^{n}(n^2 \cdot y^{13})$

Answer:

$f(n) = \sum_{y=1}^{n}(n^2 \cdot y^{13})$

$\sum_{y=1}^{n}(n^2 \cdot y^{13}) \leq \sum_{y=1}^{n}(n^2 \cdot n^{13})$

$\sum_{y=1}^{n}(n^2 \cdot n^{13}) = n^2 \cdot n^{14} = n^{16}$

$g(n) = n^{16}$

$f(n) = O(n^{16})$

$\sum_{y=1}^{n}(n^2 \cdot y^{13}) \geq \sum_{y=\frac{n}{2}}^{n}(n^2 \cdot n^{13}) \geq \sum_{y=\frac{n}{2}}^{n}(n^2 \cdot \frac{n}{2}^{13})$

$\sum_{y=\frac{n}{2}}^{n}(n^2 \cdot n^{13}) \geq$

$\sum_{y=\frac{n}{2}}^{n}\left(n^2 \cdot \left(\frac{n}{2}\right)^{13}\right) = \frac{n}{2} \cdot n^2 \cdot \frac{n^{13}}{2^{13}}$

$\frac{n}{2} \cdot n^2 \cdot \frac{n^{13}}{2^{13}} = \frac{1}{2^{14}} \cdot n^{16}$

$f(n) \geq n^{16}$

$f(n) = \Omega(n^{16})$

$f(n) = \theta(n^{16})\blacksquare$

2. Let $n \geqslant 3$. Consider an ordered list of numbers, $L = (L_1, L_2, \ldots, L_{n-1}, L_n)$

$X_k = \sum_{i=2}^{k}(L_i - L_{i-1})$

$Y = \sum_{j=3}^{n}(X_j - X_{j-1})$

Answer:

X_k and Y are telescoping series

$X_k = \sum_{i=2}^{k}(L_i - L_{i-1}) = L_k - L_1$

$Y = \sum_{j=3}^{n}(X_j - X_{j-1}) = X_n - X_2$

$X_n = L_n - L_1$

$X_2 = L_2 - L_1$

$X_n - X_2 = L_n - L_1 - (L_2 - L_1)$

$X_n - X_2 = L_n - L_1 - L_2 + L_1$

$Y = X_n - X_2 = L_n - L_2 \blacksquare$